Jinwei Liu

M.Eng. Student in Human-Robot Shared Autonomy

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Profile

- Research interests include human-machine hybrid intelligence, human-robot interaction, shared autonomy, and embodied intelligence.
- Current work focuses on shared control, teleoperation, reinforcement learning, and human-in-the-loop reinforcement learning.
- Interested in both human-to-machine intent communication and machine-to-human information expression.

Education

Master of Engineering — **University of Science and Technology of China**, Artificial Intelligence — Hefei, China — Sept 2024 – present
- Weighted average: 91.09; GPA: 3.99/4.3

Bachelor of Engineering — **Ocean University of China**, Automation — Qingdao, China — Sept 2020 – June 2024
- Rank: 1/89; Weighted average: 92.42; GPA: 3.82/4.0
- Honors: National Scholarship (twice); Honorary Bachelor's Degree of Ocean University of China

Publications

Delay-Aware Shared Control for Teleoperation Systems: Intent Trajectory Prediction Under Communication Latency — 2026
Journal article on delay-aware shared control for teleoperation systems, using DA-RT to predict operator intent under communication latency.
Jinwei Liu, Pengfei Li, Yaqing Zhou, Tao Wang, Yu Kang, Yun-Bo Zhao
ieeexplore.ieee.org/document/11535330

LSTM-MPC: Physically Feasible and Interpretable Trajectory Prediction — 2026
Accepted paper on physically feasible and interpretable trajectory prediction via latent MPC parameter estimation.
Jinwei Liu, Pengfei Li, Qianqian Zhang

Human-AI Shared Control via Motor Knowledge Neuron Stimulation — 2026
Accepted paper on human-AI shared control through distributed motor knowledge neuron assemblies.
Jinwei Liu, Pengfei Li, Sibin Huang, Yun-Bo Zhao, Yu Kang